

Review

Reactions of Plant Phenolics with Food Proteins and Enzymes under Special Consideration of Covalent Bonds

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Secondary plant metabolites are important native food components, which are becoming more and more interesting due to their physiological effects on human beings. One of the largest groups of these compounds is represented by plant phenols. This review summarizes the structure, classification and distribution of the phenolic compounds in plant foods, their chemistry and signification with regard to food processing and -storage as well as their physiological effects. This work focuses mainly on such reactions of the phenolic substances with proteins and enzymes that lead to covalent bonds. The derivatives formed have been characterized in terms of changes in their physicochemical and structural properties. The effect on the proteolytic *in vitro* digestion has been also illustrated. Further aspects reported include the influence on enzyme activity and -kinetic parameters. The different aspects of the nutritional-physiological consequences of such reactions in food and body, especially considering their significance to food science and technology are discussed.

Keywords: phenolic acids, flavonoids, food protein/enzyme derivatization, physicochemical characterisation, structure, *in-vitro* proteolytic degradation, biological activity

1. Phenols as natural substances of plant foods

1.1 Introduction

Phenolic compounds as food components represent, with more than 6000 identified substances, the largest group of secondary metabolites in plant foods. They are characterized by a large range of structures and functions, but generally possessing an aromatic ring bearing one or more hydroxy substituents (Robards *et al.*, 1999). They are usually found in plants bound to sugars as glycosides (Hollman & Arts, 2000). Plant phenols are present in many foodstuffs as colour imparting ingredients (e.g. anthocyanins in red wines, red cabbage, eggplant, strawberry, blueberry, raspberry, plum, cherry etc.) (Robards *et al.*, 1999; Clifford, 2000a; Hollman & Arts, 2000). Interest in these compounds is related to their dual role as substrates for oxidative browning reactions and as antioxidants, underlining their impact on organoleptic and nutritional qualities of fruits and vegetables, their role in plant growth and metabolism and, more recently, their demonstrated physiological activity in humans (Friedman, 1996; Robards *et al.*, 1999; Hollman & Arts, 2000). In both roles, the key process is oxidation (Robards *et al.*, 1999). The oxidative browning is probably restricted to foods and is invariably detrimental although, in some instances (e.g. tea, cacao, dates, currants, sultanas), it is intentional and essential to the character of the product (Friedman, 1996; Robards *et al.*, 1999). The action of phenolics as antioxidants is viewed as beneficial in both foods (e.g. gallates as supplementary food ingredient) and the body where phenolics are oxidized in preference to other food constituents or cellular components and tissues (Vinson *et al.*, 1995; Hertog *et al.*, 1997; Meyer *et al.*, 1998; Robards *et al.*, 1999; Yoshida *et al.*, 1999; Miura *et al.*, 2001).

1.2 Structure, classification, distribution of phenols in plant foods, intake and their physiological effects

With regard to their structure, the major groups of phenolic compounds are distinguished by a number of constitutive carbon atoms in conjunction with the basic phenolic skeleton. The food relevant groupings include firstly, the hydroxybenzoic acid derivatives (HBAs) with a general structure C_6-C_1 (e.g. salicylic and gallic acids) as illustrated in Fig. 1. Variations in their basic structure being hydroxylations and methoxylations of the aromatic ring. As a general rule HBAs are present as conjugates (Tomás-Barberán & Clifford, 2000a), although they can also be detected as free acids in some fruits (e.g. gallic acid in persimmons), after being released during fruit and vegetable maturation or more importantly during food processing. The main source of gallic acid is tea, especially green tea, where it occurs as (–)-epigallocatechin-3-gallate (EGCG) and (–)-epigallocatechin. The consumption of EGCG was estimated for Japan to be as high as 1 g/per day (Authors-group, 1999). A detailed account on the specific occurrence and content of hydroxycinnamic and hydroxybenzoic acid compounds in foods has also been reported (Herrmann, 1989; Authors-group, 1999; Clifford, 1999; Robards *et al.*, 1999; Clifford, 2000b; Tomás-Barberán & Clifford, 2000a).

The second group is that of those phenolic compounds having the general formula C_6-C_3 (hydroxycinnamic acid derivatives, Fig. 2), representing a series of trans-phenyl-3-propenoic acids differing in their ring substitution. These compounds are widely distributed as conjugates in plant material including many foods and beverages (Herrmann, 1989; Clifford, 1999; Clifford, 2000b). The commonest (Fig. 2) is caffeic acid (3,4-dihydroxycinnamic acid); ferulic (3-methoxy, 4-hydroxy), sinapic (3,5-dimethoxy, 4-hydroxy) and *p*-coumaric (4-hydroxy) also are wide-

spread (Clifford, 2000b). The best-known conjugate is 5-cafeoylquinic acid (IUPAC numbering, Fig. 2), commonly referred to as chlorogenic acid (CGA) (Clifford, 2000b). Accurate data on population-wide intakes of phenolic compounds are not available. Recent studies show, that the human dietary intake of chlorogenic acids and other cinnamates ranges from 25 mg up to 1 g/day depending upon the dietary constitution (Clifford, 1999). The content of chlorogenic acid is relatively high with 6–10% in green coffee beans, and a cup of coffee may contain up to 675 mg (Clifford, 1999).

The third and the largest group are represented by the flavonoids, with more than 4000 compounds characterized (Ader *et al.*, 2000). They are also the most widespread and diverse, built upon a C₆-C₃-C₆ flavone skeleton (Fig. 3), in which the three-carbon bridge between the phenyl groups is commonly cyclised with oxygen (Robards *et al.*, 1999). Several sub-classes of flavonoid are differentiated on the degree of unsaturation and degree of oxidation of the three-carbon segment (e.g. flavones, isoflavones, flavonols, flavanones, catechin, anthocyanins etc.). Within the various classes, further differentiation is possible, based on the number and nature of substituent groups attached to the rings (Robards *et al.*, 1999). The main dietary sources of flavonoids include fruits, vegetables as well as tea, and detailed accounts of their content have been reviewed recently (Kühnau, 1976; Cook & Samman, 1996; Böhm *et al.*, 1998; Robards *et al.*, 1999; Hollman & Arts, 2000; Miesan & Mohamed, 2001). The

estimated total intake of flavonoids in the USA was 1 g/person/day (expressed as glycosides) (Kühnau, 1976). This seems to be an overestimation, as recent reviews indicate a daily intake of only 6–60 mg/person/day (Böhm *et al.*, 1998; Hollman & Arts, 2000).

The largest number of flavonoids is represented by the sub-group of flavonols (Fig. 4), usually found in plants bound to sugars as O-glycosides (Robards *et al.*, 1999). The most important representatives of flavonols are kaempferol, quercetin and myricetin. Aglycones of flavonols do not occur in fresh plants but may be present as a result of food processing. Sugars are predominantly bound to the flavonoid nucleus via a β-glycosidic

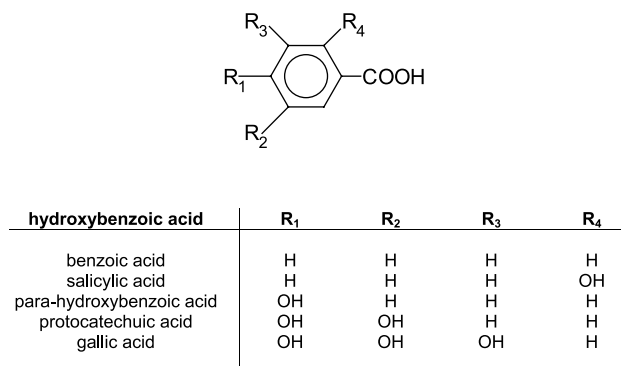


Fig. 1. The structure of hydroxybenzoic acid derivatives with a general structure C₆-C₁.

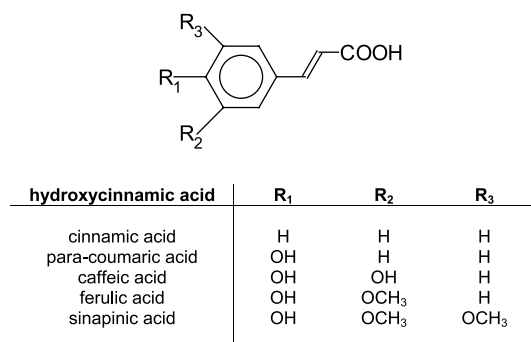


Fig. 2. The structure of hydroxycinnamic acid derivatives with a general formula C₆-C₃.

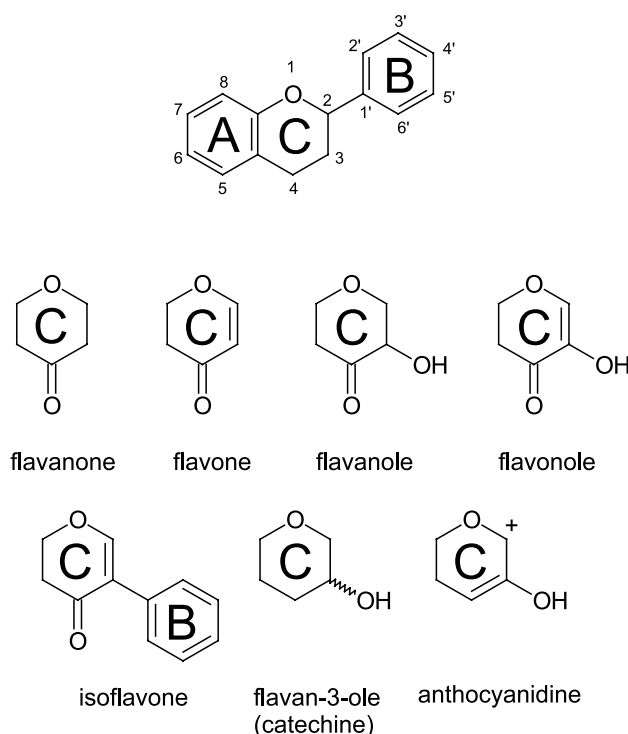


Fig. 3. The structure of the Flavonoids with a C₆-C₃-C₆ flavone skeleton.

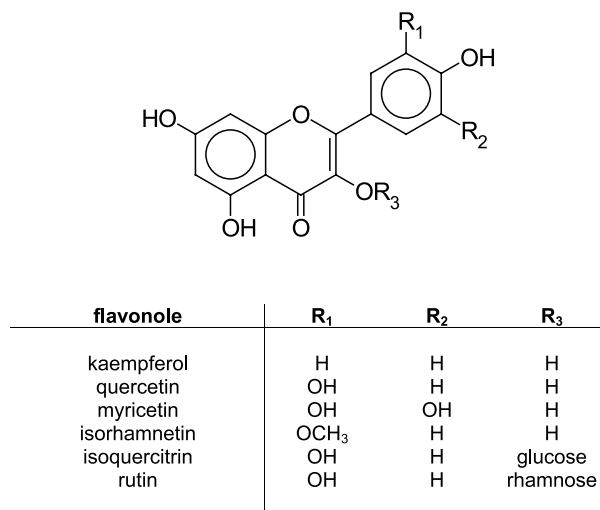
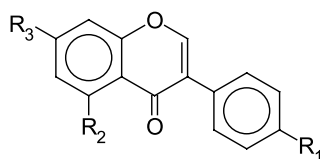


Fig. 4. The structure of the sub-group of flavonols.



isoflavone	R ₁	R ₂	R ₃	R ₄
daidzein	OH	H	OH	H
genistein	OH	OH	OH	H
glycitein	OH	H	OH	OCH ₃
prunetin	OH	OH	OCH ₃	H
formonetin	OCH ₃	H	OH	H
biochanin A	OCH ₃	OH	OH	H

Fig. 5. The structure of the sub-group of isoflavones.

bond. Sugar molecules can bind to various positions in the parent flavonoid, although there is a preference for the C₃-position. 179 different glycosides of quercetin alone have been described in nature (Hollman & Arts, 2000), rutin (quercetin-3-O-rhamnosyl-glucoside) being one of the commonest. A further physiologically interesting sub-group of the flavonoids is that of isoflavones (Fig. 5), representing the hormone-like bisphenolic phytoestrogens of dietary origin (Mazur *et al.*, 1998; Cassidy *et al.*, 2000; Ren *et al.*, 2001). More than 60 representatives of this class have been identified in plants, especially those of leguminous seeds, soy beans being the most important source and the commonest isoflavones being genistein and daidzein (Cassidy *et al.*, 2000).

Intense research activity in recent years has been directed towards the absorption, metabolism and bioavailability of phenolic compounds and detailed data on the bioavailability and uptake of selected phenolic compounds has been reported (Cook & Samman, 1996; Hollman *et al.*, 1996; Hollman *et al.*, 1997; Böhm *et al.*, 1998; Bourne & Rice-Evans, 1998; Lapidot *et al.*, 1998; de Vries *et al.*, 1998; Chesson *et al.*, 1999; Clifford, 1999; Hollman *et al.*, 1999; Ader *et al.*, 2000; Andlauer *et al.*, 2000; Azuma *et al.*, 2000; Cassidy *et al.*, 2000; Clifford, 2000a,b; Clifford & Scalbert, 2000; Hollman & Arts, 2000; Tomás-Barberán & Clifford, 2000a,b; Andlauer *et al.*, 2001; Baba *et al.*, 2001; Cao *et al.*, 2001; Olthof *et al.*, 2001; Adam *et al.*, 2002).

The flavonoids have been reported to exhibit a wide range of positive biological effects including anticarcinogen (Brown, 1980; Stich & Rosin, 1984; Huber *et al.*, 1997; Mazur *et al.*, 1998; Manthey & Guthrie, 2002), antimutagen (Stich & Rosin, 1984; Malaveille *et al.*, 1996), antibacterial (Cook & Samman, 1996), antiviral (Cook & Samman, 1996), anti-inflammatory (Cook & Samman, 1996), antiallergic (Cook & Samman, 1996; Noguchi *et al.*, 1999), antithrombic and vasodilatory effects (Gryglewski *et al.*, 1987; Middleton & Kandaswami, 1992; Cook & Samman, 1996). Most of these positive effects result from their property as antioxidants, free radical scavengers and chelators of divalent cations (Cook & Samman, 1996; Rice-Evans *et al.*, 1996; Meyer *et al.*, 1998; Robards *et al.*, 1999; Yoshida *et al.*, 1999; Miura *et al.*, 2001; Hodek *et al.*, 2002). The phytoestrogens and their metabolites have many potent hormonal and non-hormonal activities that may explain some of the biological effects of diets rich in isoflavones (Mazur *et al.*, 1998; Setchell, 1998; Song *et al.*, 1999; Ren *et al.*, 2001; Arjmandi & Smith,

2002). On the other hand, depending on the concentration and hydroxyl substituent pattern, phenolic compounds can also act as pro-oxidants (Laughton *et al.*, 1989; Yamanaka *et al.*, 1997; Metodiewa *et al.*, 1999). Other negative physiological effects of flavonoids (mutagenicity, genotoxicity, cytotoxicity) have also been described (MacGregor & Jurd, 1978; Brown & Dietrich, 1979; Brown, 1980; Rueff *et al.*, 1992; Yamanaka *et al.*, 1997; Metodiewa *et al.*, 1999; Hodek *et al.*, 2002). Most of these physiological effects are based on either *in vitro* models, cell/tissue culture studies, animal experiments or epidemiological data. The results thus obtained are generally projected to show if a strong association or correlation of activity in humans is possible or not, since data on basis of human investigations are relatively rare.

1.3 Chemistry and signification regarding to food processing and -storage

Enzymatic browning is the most known reaction of plant foods. Browning reactions can be in some instances (e.g. tea, cacao, dates, currants, sultanas), intentional and essential to the character (e.g. coloring and taste) of the product (Friedman, 1996; Robards *et al.*, 1999). The action of phenolics as antioxidants is perhaps the most beneficial effect in both foods and the body (Cook & Samman, 1996; Rice-Evans *et al.*, 1996; Meyer *et al.*, 1998; Robards *et al.*, 1999; Yoshida *et al.*, 1999; Miura *et al.*, 2001; Hodek *et al.*, 2002). In order to function as an antioxidant, the phenolic compound must be capable of forming a resonance stabilized phenoxyl radical, which in turn can react with other available radicals (e.g. peroxy radical), consequently deactivating them (Robards *et al.*, 1999; Pedrielli *et al.*, 2001). However, the reactivity of the phenoxyl radicals should remain so low, such that the radical formation e.g. by H-subtraction from unsaturated fatty acids is not possible. A further beneficial property of phenolic compounds in food processing and -storage is their antimicrobial ability (Scalbert, 1991; Cook & Samman, 1996).

The presence of polyphenols in foods and beverages (e.g. sorghum, millet, fruit, cocoa, coffee, tea, beer, and wine) also produces a sensation of astringency, which is perceived as a diffuse feeling of extreme dryness and roughness that is not confined to a particular region of the mouth or tongue, thus effecting the taste of foodstuffs (Baxter *et al.*, 1997). Proteins and polyphenolic compounds can combine to form soluble complexes; these can grow to colloidal size, leading to visible haze formation and as a result limiting the shelf life of products the consumer expects to be clear including beer, wine, fruit juices, coffee and tea (Siebert *et al.*, 1996; Baxter *et al.*, 1997; Siebert, 1999). The same phenomenon can contribute to the fouling of process equipment surfaces with deposits that are difficult to remove by in-place cleaning (Siebert *et al.*, 1996). Polyphenol-oxidase-catalyzed oxidation of polyphenol compounds in fruits and vegetables to quinones, which then polymerize to dark melanin pigments of unknown structure causing discoloring of foodstuffs, e.g., formation of brown or black spots from chlorogenic acid in bananas, pears, lettuce, and potatoes and the browning of fruit juices (Friedman, 1996). Enzyme-catalyzed browning reactions of amino acids and proteins with oxidized plant phenols may cause deterioration of food during storage and processing leading to a loss in nutritional quality, which is especially serious in underprivileged countries (Friedman, 1996). Plant phenols may also display many other possible detrimental effects, including inhibition of iron absorp-

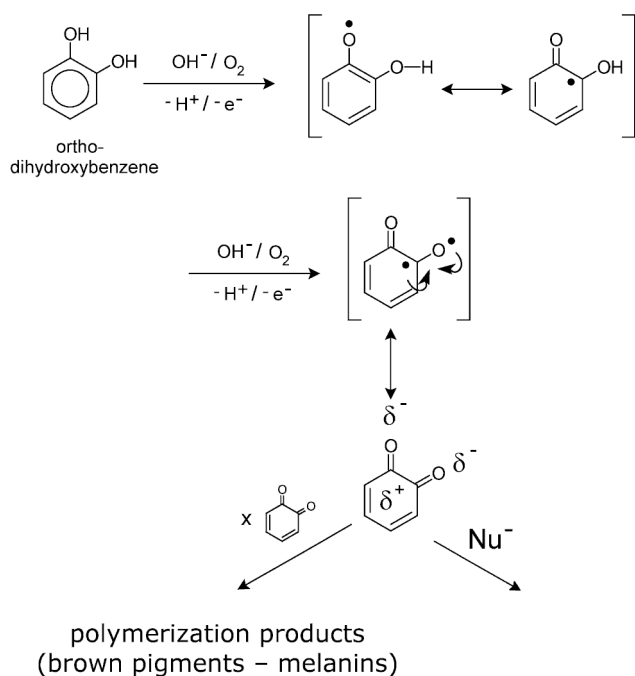


Fig. 6. The basic reactions determining the chemical reactivity of phenolic compounds.

tion and irreversible complexation of dietary proteins (Mehansho *et al.*, 1987; Baxter *et al.*, 1997). The loss of nutritional quality is also attributed to the destruction of essential amino acids and a decrease in digestibility and inhibition of proteolytic and glycolytic enzymes (Friedman, 1996). The production of antinutritional and toxic compounds may further reduce the nutritional value and possibly the safety of foods (Friedman, 1996).

With regard to the chemical reactivity of phenolic compounds, it is now generally accepted that two reaction steps are involved (Nicolas *et al.*, 1994). The first reaction step consists of the hydroxylation of monophenols into *o*-diphenols and the second, the oxidation of the *o*-diphenols into *o*-quinones. These two reactions consume oxygen and are catalyzed by the enzymes monophenolase (E.C. 1.14.18.1, cresolase activity) and *o*-diphenolase (E.C. 1.10.3.1, catecholase activity) (Nicolas *et al.*, 1994). Similarly the laccases (E.C. 1.10.3.2) oxidize *o*-diphenols as well as *p*-diphenols forming their corresponding quinones. The quinones resulting as the primary products of enzymatic oxidation represent a species of highly reactive substances (Fig. 6) which normally react further with other quinones to produce dark melanin pigments (Nicolas *et al.*, 1994; Friedman, 1996; Robards *et al.*, 1999). The quinone, being a reactive electrophilic intermediate can also readily undergo attack by nucleophiles (Fig. 6) such as lysine, methionine, cysteine and tryptophan residues in a protein chain, leading to protein derivatization and consequently causing changes in their physicochemical, structural and biological properties.

2. Reactions of phenolics and related substances with food proteins

Most of the literature sources deal with non-covalent type of interactions between phenolic compounds and proteins. There is only a very limited, especially experimental data on the covalent

bonds between these two reactants. Principally, five potential types of interactions of phenolics and proteins can be proposed: hydrogen bonding, π -bonding, and hydrophobic, ionic, and covalent linkages (Hagerman, 1992; Bianco *et al.*, 1997). The main mechanisms of protein polyphenol interaction at moderate pH are thought to be governed by π -bonding (Bianco *et al.*, 1997). Phenols may also be oxidized with ease in an alkaline solution to its corresponding quinone (Hurrell & Finot, 1984). The quinone, being a reactive electrophilic intermediate, can readily undergo attack by nucleophiles such as lysine, methionine, cysteine and tryptophan moieties in a protein chain (see also chapter 1.3) (Pierpoint, 1969a,b; Hurrell *et al.*, 1982; Hurrell & Finot, 1984; Matheis & Whitaker, 1984a,b; Machholz & Lewerenz, 1989; Nicolas *et al.*, 1994). Upon further oxidation of this addition product to form its quinone, a second addition may occur, which leads to formation of cross-linked protein polymers (Matheis & Whitaker, 1984a,b; Machholz & Lewerenz, 1989). In most of these reports, theoretical aspects of covalent linkages between proteins and phenolic compounds with corresponding reaction schemes are discussed. Few results showing such reactions of *o*-quinone with proteins are reported (Pierpoint, 1969a,b; Vithayathil & Murthy, 1972; Hurrell *et al.*, 1982; Hurrell & Finot, 1984; Muralidhara & Prakash, 1995).

Contrary to the above-mentioned non-covalent type of interactions, our main interest was to focus our studies on covalent linkages between plant phenols and proteins/enzymes. Recently, it was possible for the authors of this review to derivatize proteins/enzymes with phenols and related compounds in absence of polyphenoloxidases. The derivatives were characterized, to demonstrate the changes induced in their physicochemical and structural properties and as well as in the biological activities of the enzymes. The food proteins and enzymes tested by us included myoglobin, lysozyme, bovine serum albumin, whey proteins, soy proteins, α -amylase, bromelain, chymotrypsin and trypsin. A series of simple phenolic and related compounds (quinic-, ferulic-, caffeic-, chlorogenic-, and gallic acid; resorcinol, pyrocatechol, hydroquinone representing *m*-, *o*-, *p*-dihydroxyphenols and finally *p*-quinone) were applied and the results reported in detail (Kroll *et al.*, 2000; Rawel *et al.*, 2000a,b; Kroll *et al.*, 2001; Kroll & Rawel, 2001; Rawel *et al.*, 2001a,b; Rohn *et al.*, 2001; Rawel *et al.*, 2002a,b; Rohn *et al.*, 2002a,b). The studies were further extended to include the reaction of proteins with flavonoids (flavone, apigenin, kaempferol, quercetin and myricetin; $\text{C}_6\text{-C}_3\text{-C}_6$ structure) (Kroll *et al.*, 2002; Rawel *et al.*, 2002a; Rawel *et al.*, 2003). Some of the characteristic changes in proteins and enzymes derivatized with these phenolic compounds are discussed with help of recent and partly unpublished data in the following sections.

2.1 Reactions at the reactive side chains of proteins: amino-, sulfhydryl groups and tryptophan

The derivatization of the proteins was conducted under relatively mild conditions (pH 5–9, room temperature, different concentrations) as found during the technological food processing. The specific details can be referred to in our publications listed above.

The amount of free amino groups was determined using trinitrobenzenesulfonic acid (TNBS), which binds to the nucleophilic nitrogen of the ϵ -amino side chains of lysine. The reaction of the

phenolic compounds with the free amino groups of the proteins results in the consequent binding and blocking of this reaction site for TNBS. This leads to a corresponding decrease in the amount of the free amino groups in the myoglobin derivatives of flavonoids as illustrated in Fig. 7. Since the amount of free amino groups was determined in presence of 1% SDS (a well known agent, which destroys non-covalent protein interactions), we can assume, that the derivatization has occurred through covalent binding. The reaction of lysine side chains of myoglobin with simple phenolic and related compounds (quinic-, ferulic-, caffeic-, chlorogenic-, and gallic acid, *m*-, *o*-, *p*-dihydroxyphenols and *p*-quinone) has already been reported (Kroll *et al.*, 2000; Kroll & Rawel, 2001). The basic requirement for such a reaction is the formation of a corresponding quinone, this condition being fulfilled by caffeic-, chlorogenic-, and gallic acid, *o*- and *p*-dihydroxyphenols. Ferulic acid and *m*-dihydroxyphenol cannot be oxidized to their corresponding quinone derivative, but were found to react with the proteins applied (Rawel *et al.*, 2000b). One possible explanation can be given by considering the formation of a radical, which in turn can attack a nucleophilic reaction partner. The possible reaction intermediates from ferulic acid and *m*-dihydroxyphenol have been postulated (Kroll & Rawel, 2001; Rawel *et al.*, 2001b). Quinic acid as a simple cyclic compound also reacts with the free amino groups only to a small, but significant level (Rawel *et al.*, 2000b). The mechanisms of this reaction are not yet clear.

From the flavonoids tested, flavone (having no hydroxy substituents) and apigenin (hydroxy substituents on ring A and 4' of ring B) showed no significant reactivity towards the free amino groups, whereas kaempferol, quercetin and myricetin did undergo a reaction (Fig. 7). This would mean for quercetin and myricetin that primarily the hydroxy substituents on the ring B and C (positions 3, 3', 4', 5') of the flavonoids invoke a strong reaction with lysine side chains in proteins. A possible route for the reaction of kaempferol with proteins can be explained by means of enol-keto tautomerization across the 2–3 bond (Fig. 8). Therefore, the occurrence of the catechol moiety, i.e. the two adjacent (ortho) aromatic hydroxyl groups in quercetin and myricetin (C3' and C4', ring B) and the subsequent autoxidation

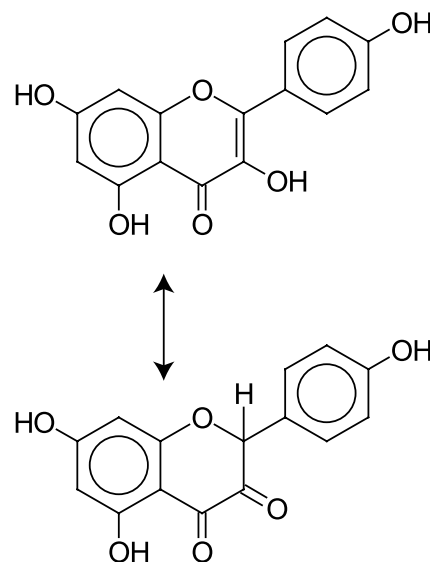


Fig. 8. Enol-keto tautomerization in kaempferol.

(alkaline pH 9, presence of oxygen during the derivatisation) of these to reactive and redox active *o*-quinones appears to be a prerequisite condition, to generate electrophilic species capable of undergoing such nucleophilic addition.

A further possibility of reaction of oxidized phenolic compounds is at the heterocyclic *N*-atom of tryptophan as already discussed (Matheis & Whitaker, 1984a; Machholz & Lewerenz, 1989). The quenching of the fluorescence intensity as an indicator of tryptophan changes was studied after activation at 295 nm and measurement of the emission between 300–360 nm. The results did also show a decrease of tryptophan content in dependence of the added reactant as documented in Fig. 9 for the applied flavonoids. A similar observation was also made with simple phenolic and related compounds as reported earlier (Kroll *et al.*, 2000; Rawel *et al.*, 2000a,b; Kroll & Rawel, 2001; Rawel *et al.*, 2001a,b). Further, from the flavonoids (Fig. 9), even flavone and apigenin were found to react. Kaempferol, quercetin as

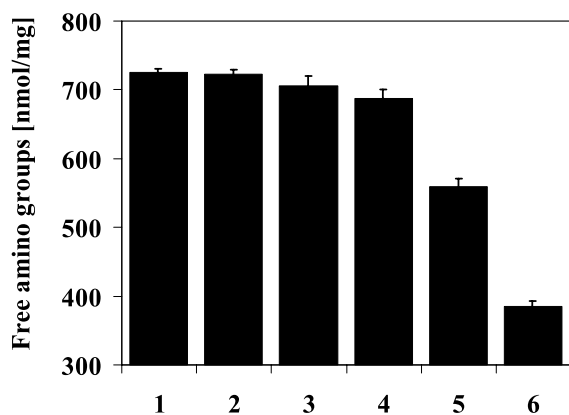


Fig. 7. Changes in the amount of the free amino groups in the myoglobin derivatives of flavonoids (1.07 mmol phenol/g protein). Code: 1=unmodified (control) myoglobin; 2=flavone derivative; 3=apigenin derivative; 4=kaempferol derivative; 5=quercetin derivative; 6=myricetin derivative.

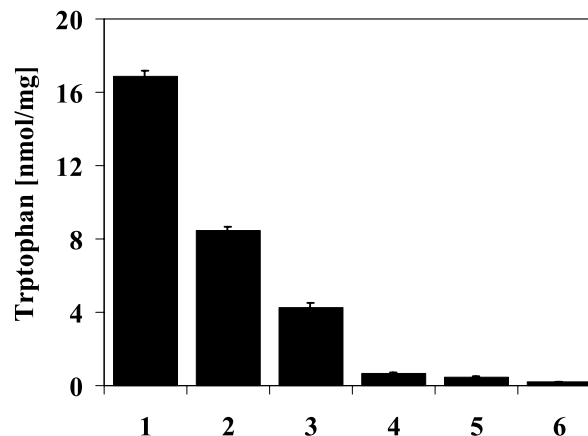


Fig. 9. The change of tryptophan content of myoglobin in dependence of the added reactant (1.07 mmol phenol/g protein). Code: 1=unmodified (control) myoglobin; 2=flavone derivative; 3=apigenin derivative; 4=kaempferol derivative; 5=quercetin derivative; 6=myricetin derivative.

well as myricetin were more reactive (Fig. 9). The mechanisms of the interactions of flavone with the tryptophan groups are not yet clear and that of apigenin are likely to be similar to that with ferulic acid over the formation of a radical as mentioned above. Since the tryptophan fluorescence was measured in presence of 8 M urea (destroying of non-covalent protein interactions), it can be assumed that covalent binding has occurred. These trends in the decrease of tryptophan were also confirmed by performing alkali hydrolysis of the derivatives, followed by consequent amino acid analysis (unpublished data).

Besides the above mentioned influence on lysine and tryptophan concentration of proteins, the content of the thiol groups can also be decreased by undergoing similar reactions with oxidized phenolic compounds as postulated and documented (Rawel *et al.*, 2000b; Kroll *et al.*, 2002; Rawel *et al.*, 2002a,b). As recently reported, *o*-quinone trapping method by glutathione appeared to be an accurate method for quantification of *o*-quinone metabolites (Awad *et al.*, 2000; Boersma *et al.*, 2000; Awad *et al.*, 2001; Awad *et al.*, 2002a,b). In these studies, the method proved useful for analysing the nature and biochemical behavior of quercetin *o*-quinone, and characterization of its *p*-quinone methide isomers (Awad *et al.*, 2000; Boersma *et al.*, 2000; Awad *et al.*, 2001; Awad *et al.*, 2002a,b). According to these authors, for quercetin *o*-quinone and *p*-quinone methide III, glutathione adducts can be formed only at C2', C5', and C6', whereas for *p*-quinone methides I and II, adduct formation at C2', C5', C6', C6, and C8 is possible (Boersma *et al.*, 2000; Awad *et al.*, 2001). Translating this information to our case and consequently in accordance to the observed decrease in the nucleophilic protein side chains, similar covalent adduct formation at the proposed C atoms for the flavonoids is possible.

This estimation of the reactivity of the protein side chains is incomplete, since it has been theoretically and experimentally observed that further amino acid residues like proline, methionine, histidine and tyrosine also have the potential to react with the phenolic compounds (Matheis & Whitaker, 1984a; Machholz & Lewerenz, 1989; Rawel *et al.*, 2000b).

On the other hand, the methods of characterization and identification of plant phenolics show that their spectrophotometric absorption properties can be applied to estimate the amount of the phenolic compound covalently bound to the proteins. But consideration should be given to the fact that the extinction coefficient and absorption maximum wavelength can change in an adducted species. Hydroxycinnamic acids (like chlorogenic acid) and flavonoids (like quercetin) exhibit absorption bands at 305–330 nm and 320–385 nm respectively (Robards *et al.*, 1999). This absorption characteristics were successfully applied to estimate the amount of the individual chlorogenic acid and quercetin covalently bound (Rawel *et al.*, 2002b; Rawel *et al.*, 2003).

2.2 Changes in solubility induced by derivatization

Solubility is a prime requisite for a functional ingredient protein and is critically necessary for products such as beverages. For example, whey proteins are nitrogenous fraction remaining soluble in the supernatant at pH 4.6 after precipitation of casein. Thus, the loss in solubility at this pH is commonly used to access the extent of protein denaturation (Kinsella & Whitehead, 1989). Therefore, it was possible to estimate the degree of denaturation in whey proteins caused by their reaction with the simple phenol-

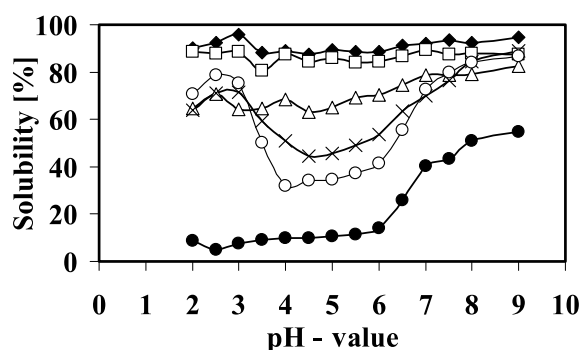


Fig. 10. The Solubility behaviour of myoglobin-flavonoid derivatives (1.07 mmol phenol/g protein). Code: 1=◆ unmodified (control) myoglobin; 2=□ flavone derivative; 3=△ apigenin derivative; 4=× kaempferol derivative; 5=○ quercetin derivative; 6=● myricetin derivative.

ic and related compounds (Rawel *et al.*, 2001a) and recently with flavonoids (Rawel *et al.*, 2003). The complexation of myoglobin with oxidized flavonoids is demonstrated by the pH dependent solubility profile, whereby the unmodified control myoglobin had a good solubility in buffer at the investigated pH range (Fig. 10). The reaction of myoglobin with the derivatizing flavonoids is accompanied by a corresponding blocking of the hydrophilic amino acid side chains, causing a parallel change of its solubility (Fig. 10). Flavone caused hardly any influence on the solubility behavior, whereas apigenin, kaempferol, quercetin as well as myricetin produced significant changes (Fig. 10). These changes in solubility reflect the respective changes in the amount of charged groups as documented for free available amino groups in Fig. 7, having a good correlation with the structure of the flavonoids. Certainly, with pH near 4, phenols are also known to complex with proline-containing proteins by non-covalent associations that lead to reduced solubility (Siebert, 1999). Myoglobin has about 2.5 mol% proline (GenBank Protein sequences, Accession No. 2506462 for myoglobin from horse). Under these conditions a complexation of phenols with proline would be expected, whereby phenolic compounds are known to be much stronger in binding to proline-rich proteins than other configurations (Siebert & Lynn, 1998; Siebert, 1999). From the nutritional and technological-functional points of view, such decrease in solubility may adversely affect not only protein extraction from raw materials with a relative high content of phenolic compounds (such as sorghum, sunflower seeds, rapeseed), but also the digestion properties as shown below.

2.3 Changes in hydrophilic/hydrophobic character of the protein derivatives

The surface activity of proteins is important in a number of functional applications, e.g. foams and emulsions. The number and disposition of apolar groups contribute to the effective hydrophobicity of proteins by promoting a greater affinity of the protein e.g. for an oil phase. The covalent attachment of the phenolic compound to proteins causes on one hand the blocking of the hydrophilic groups like amino and thiol groups as discussed above, but on the other hand there is also an increase in the amount of apolar groups (benzene ring) and polar groups (hydroxyl and carboxyl groups as in the case of the phenolic acids)

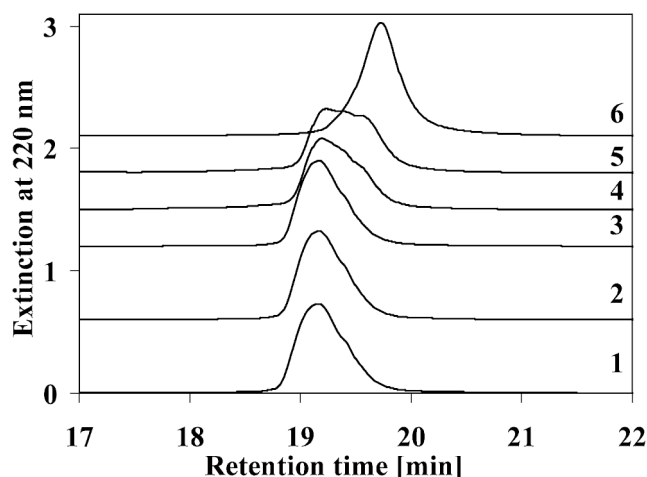


Fig. 11. RP-HPLC of the myoglobin-flavonoid derivatives (1.07 mmol phenol/g protein). Code: 1=unmodified (control) myoglobin; 2=flavone derivative; 3=apigenin derivative; 4=kaempferol derivative; 5=quercetin derivative; 6=myricetin derivative.

being introduced. Simultaneously, the solubility behaviour indicates denaturation, which means structural and conformational changes. All these changes affect the hydrophilic/hydrophobic character of the derivatized proteins, which can be illustrated by RP-HPLC. This is documented in a rise of the retention times of the main peak e.g. illustrated for myoglobin flavonoid derivatives in Fig. 11. The peak formation may be adversely affected, indicating possible denaturation and molecular interactions (Fig. 11). Contrary to these results, an increase in the hydrophilic character (using RP-HPLC) of the derivatives of whey proteins with phenolic compounds was also observed (Rawel *et al.*, 2001a; Rawel *et al.*, 2003). To summarize, it appears that the effect of the treatment of proteins with phenolic compounds on their hydrophilic/hydrophobic character seems to be governed by different factors, some of which like pH, phenol concentration, protein and/or phenol configuration have been documented (Kroll *et al.*, 2000; Rawel *et al.*, 2000a,b; Kroll *et al.*, 2001; Kroll & Rawel, 2001; Rawel *et al.*, 2001a,b; Rohn *et al.*, 2001; Rawel *et al.*, 2002a,b; Rawel *et al.*, 2003). On the other hand, to optimize the biological and technological utilization of proteins, especially of low-quality proteins in underdeveloped countries, a better understanding is needed of the various interrelated parameters that influence their nutritional value and functional properties. As noted above, the hydrophobic-hydrophilic properties and solubility behavior have also been modified by reactions of proteins with phenolic compounds, which in turn may influence the functional properties (emulsion-, foam properties) of the derivatives.

2.4 Characterization of the protein-phenol-derivatives with electrophoretical methods

As a result of the covalent linkages of the phenolic compounds with the proteins and in accordance with the observed changes in the pH dependent solubility of the derivatives (Fig. 10), there is a change in the net charge of the protein molecules, which is reflected in the corresponding change of the isoelectric point of the derivatives. Exemplary, the changes obtained by the reaction of selected flavonoids with myoglobin is depicted in the Fig. 12. With the increasing number of hydroxyl groups and their posi-

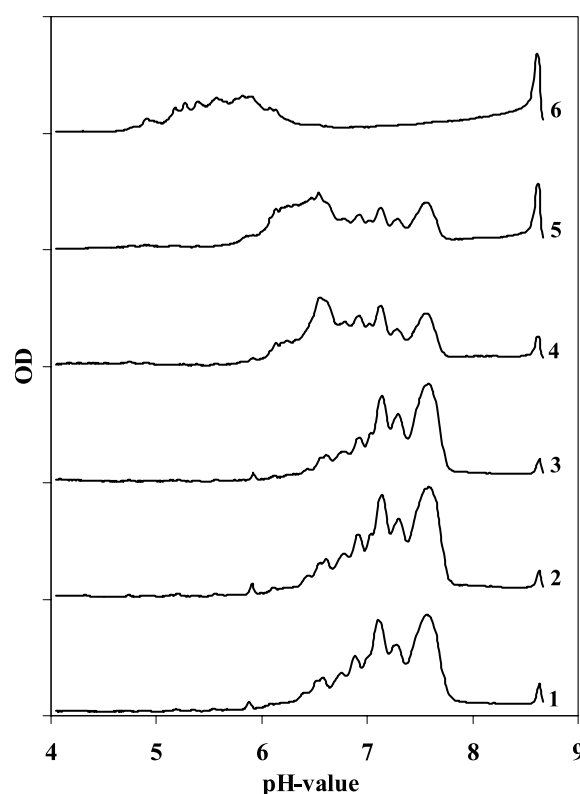


Fig. 12. Isoelectric focussing (IEF) of myoglobin-flavonoid derivatives (1.07 mmol phenol/g protein). Code: 1=unmodified (control) myoglobin; 2=flavone derivative; 3=apigenin derivative; 4=kaempferol derivative; 5=quercetin derivative; 6=myricetin derivative.

tions in the reacting flavonoids and their corresponding higher reactivity as witnessed by the blocking of free amino groups (Fig. 7) and tryptophan (Fig. 9), a shift in the isoelectric points of the myoglobin fractions can be documented (Fig. 12). Generally we can summarize following: With higher degrees of derivatization, lower isoelectrical points for proteins can be observed i.e. the proteins become more acidic in nature. This observation is consistent in all our investigations regarding interactions of proteins with phenolic compounds.

The estimated data obtained for the total number of amino acid residues blocked in proteins (BSA, whey proteins) was generally higher as compared to the amount of phenolic compound bound (chlorogenic acid, quercetin) (Rawel *et al.*, 2002b; Rawel *et al.*, 2003) underlining the fact that some protein groups must be involved in intra- and intermolecular cross-linking. The later can be confirmed by SDS-PAGE analysis as documented in Fig. 13 for myoglobin-flavonoid derivatives. Again flavon and apigenin did not show any change in the electrophoretical behavior, whereas kaempferol, quercetin as well as myricetin induced strong polymerization. These high molecular complexes produced, were not spliced by use of SDS and mercaptoethanol. The latter reagent is generally used to split disulfide bonds during sample preparation. This is a further indication, that the bonds formed between the molecules must be of covalent nature. A summary of the nature of several of the reaction products reported in the literature to be formed upon the chemical or enzymatic oxidation of flavonoids is given in Awad *et al.* (2000). The general picture emerging from these studies is that the oxidation of fla-

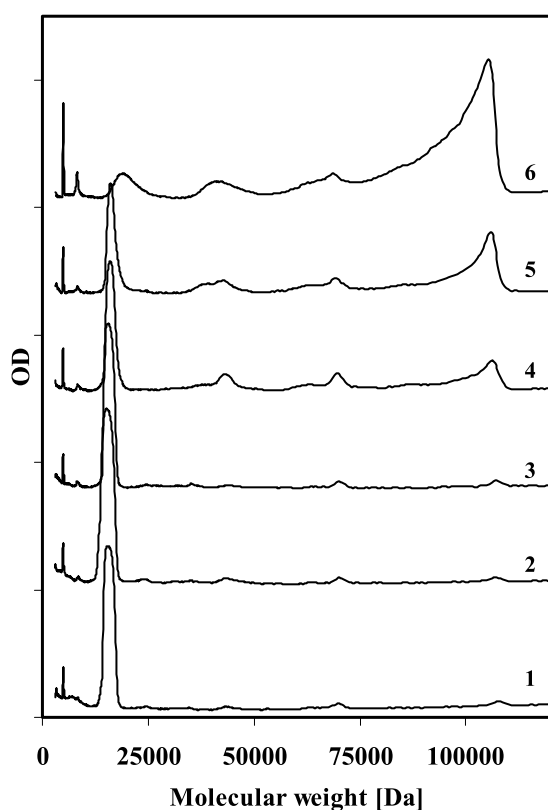


Fig. 13. SDS-PAGE of the myoglobin-flavonoid derivatives (1.07 mmol phenol/g protein). Code: 1=unmodified (control) myoglobin; 2=flavone derivative; 3=apigenin derivative; 4=kaempferol derivative; 5=quercetin derivative; 6=myricetin derivative.

vonoids is complex, resulting in a wide variety of, in many cases, unidentified reaction products. In the presence of glutathione, however, these products were no longer observed, formation and dimerization of glutathione adducts originating from glutathione conjugation with quercetin were observed and identified by ^1H NMR (Awad *et al.*, 2000; Boersma *et al.*, 2000; Awad *et al.*, 2001; Awad *et al.*, 2002a,b). According to these studies, the adduct formation at following sites-C2', C5', C6', C6, and C8 of quercetin is possible.

2.5 Characterization of the protein-phenol-derivatives with mass spectrometry

The confirmation of the covalent addition of the phenolic compounds to the monomer protein molecules was obtained by applying the analytical methods of mass spectrometry like MALDI-TOF-MS and SELDI-TOF-MS. Exemplary, the results obtained for myoglobin with simple phenolic and related compounds (quinic-, ferulic-, caffeic-, chlorogenic-, and gallic acid, *m*-, *o*-, *p*- dihydroxyphenols and *p*-quinone) document this covalent attachment (Kroll *et al.*, 2000; Kroll & Rawel, 2001). Recent experiments with whey proteins and quercetin also confirm the covalent linkages for flavonoids (Rawel *et al.*, 2003). The mass spectra of the myoglobin derivatives with chlorogenic acid and *p*-quinone illustrating such changes are shown in Fig. 14. The reaction of myoglobin with chlorogenic acid (352.3 Da) delivered the highest molecular mass of 18025 Da, which in turn could account for addition of at least 3 molecules of chlorogeno-

quinone to one myoglobin molecule (Fig. 14). In case of *p*-quinone (108.1 Da), a high molecular weight of 18117 Da was derived, which accounts for a corresponding addition of ca.11 molecules of *p*-quinone (Kroll & Rawel, 2001).

2.6 Influence of the reaction parameters and conditions

The pre-requisite condition for the reaction of phenolic compounds with proteins is the presence of oxygen, in order to produce a reactive and redox active quinone or a semi-quinone radical, an electrophilic species capable of undergoing nucleophilic addition at the reactive side chains of the proteins. It is clear that the reaction conditions like time period, temperature, pH, oxygenation, concentration of the phenolic compounds as well as the position and the amount of hydroxyl groups present, may greatly influence the extent of their covalent attachment to proteins. The presence or absence of enzymes (e.g. tyrosinase) can also be important with regard to the oxidation products formed and consequently the reaction products produced with proteins (Hurrell *et al.*, 1982; Hurrell & Finot, 1984). Some interesting aspects of reactions of caffeic- and chlorogenic acids with casein, showing the influence of pH, reaction time, temperature and presence or absence of enzymes (tyrosinase) have been reported (Hurrell *et al.*, 1982; Hurrell & Finot, 1984). The influence of pH was also documented for reaction of chlorogenic acid with lysozyme and α -amylase, where even at low pH values (pH 4–5) derivatization occurred and consequent change of their activity was witnessed (Rawel *et al.*, 2002b; Rohn *et al.*, 2002b). With regard to flavonoids, recent results showed that, the adduct formation of quercetin quinone or quercetin quinone methide was found to be pH dependent (Awad *et al.*, 2002a). A shift in adduct formation was noted from the C ring at pH 3.5, to the A ring at pH 7.0 and finally to the B ring at pH 9.5. This indicates a significant influence of the pH on the regioselectivity of glutathione conjugation to quercetin quinone/quinone methide (Awad *et al.*, 2002a). The concentration of the phenolic compounds present during the derivatization also showed considerable effect, the degree of derivatization increasing with their rising concentration as documented for myoglobin and phenolic acids in (Kroll *et al.*, 2000) and for α -amylase activity in (Rohn *et al.*,

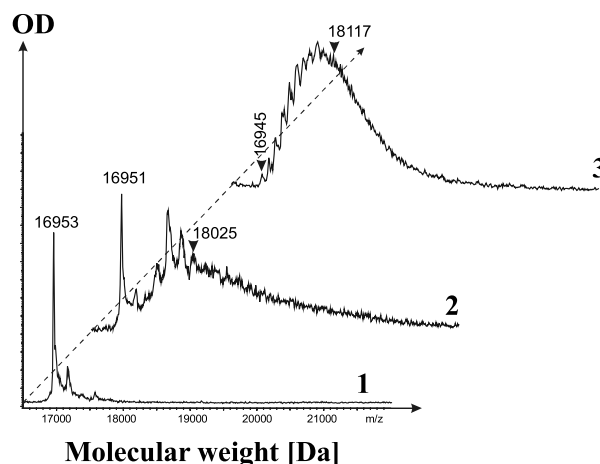


Fig. 14. MALDI-TOF-MS of myoglobin-phenol derivatives (0.42 mmol phenol/g protein). Code: 1=unmodified (control) myoglobin; 2=chlorogenic acid derivative; 3=*p*-quinone derivative.

2002b). The different configurations of hydroxyl groups, i.e. their amount and their position in the phenolic compounds, also determine the degree of protein derivatization (Kroll & Rawel, 2001; Rawel *et al.*, 2001a,b; Rawel *et al.*, 2002a; Rawel *et al.*, 2003). The dependence of the reactivity on the different configurations of hydroxyl groups in flavonoids is also demonstrated by studies using the glutathione trapping method followed by HPLC, NMR, MALDI-TOF, and LC/MS analysis to identify the different glutathionyl adducts formed (Awad *et al.*, 2000; Boersma *et al.*, 2000; Awad *et al.*, 2001).

3. Characterization of structural changes in protein-phenol-derivatives

The structural changes were studied using circular dichroism, differential scanning calorimetry, intrinsic fluorescence, and binding of anilidonaphthalenesulfonic acid. These results have been published recently (Rawel *et al.*, 2002a,b; Rawel *et al.*, 2003). A far-UV-CD study was conducted e.g. to evaluate the influence of derivatization by chlorogenic acid on the secondary structure of BSA (Rawel *et al.*, 2002b). The results showed that there was a significant perturbation of the secondary structure with increasing degree of derivatization, causing a loss of the α -helix and a parallel increase in the content of unordered structure elements (Rawel *et al.*, 2002b). The derivatisation of β -lactoglobulin with quercetin and rutin (quercetin-3-O-rhamnosylglucoside) also caused a decrease in α -helix (esp. the regular α -helix) with a parallel increase in the remaining structure confirming the destructuring effect (Rawel *et al.*, 2003). The perturbation of the secondary structure became more significant after derivatisation with rutin leading to an increased amount of unordered structure elements (Rawel *et al.*, 2003). The reason being its bulkiness resulting from the attached rhamnosylglucoside at C₃. Investigations on thermodynamics of interactions of caffeic acid and quinic acid with multisubunit proteins from sunflower seeds also suggested a marginal increase in β -sheet content, while α -helix and β -turn contents were marginally decreased (Suryaprakash *et al.*, 2000). Interaction of chlorogenic acid with human serum albumin was also shown to cause a decrease in α -helix structure with other structures unaltered (Muralidhara & Prakash, 1995). These results are in agreement with our observations for BSA and β -lactoglobulin.

The interactions of flavonoids (apigenin, kaempferol and quercetin) with soy glycinin on the other side produced an increase in α -helix and β -strand (Rawel *et al.*, 2002a). A similar observation was also made during experiments concerning interactions of a globular protein (arachin) exposed to simple polyphenols (catechol and pyrogallol) and was attributed to the number of the introduced hydroxyl groups (Neucere *et al.*, 1978). The interaction was thought to be of the hydrogen bond type, either through interpeptide chains or encompassing the phenols themselves (Neucere *et al.*, 1978). Derivatization with myricetin causes a decrease in α -helix, with a parallel increase in β -strand and -turn. In summary, CD studies and estimation of changes in secondary structures clearly demonstrates a perturbation, that is dependent of phenolic compounds and proteins applied.

The influence of the derivatization on the tertiary structure of soy glycinin and β -lactoglobulin were evaluated by means of a near-UV-CD study (Rawel *et al.*, 2002a; Rawel *et al.*, 2003). The CD of proteins in the near UV (310–240 nm) derives from transi-

tions in the prosthetic groups e.g. tryptophan. The variation of CD in this region can be used to monitor changes in conformation and local environment. Chlorogenic acid caused a decrease in the molecular ellipticity compared to the unmodified soy glycinin, where as caffeic- and gallic acid an increase. Conversely, flavonoids-flavone and apigenin caused hardly any change, kaempferol only a slight, and quercetin as well as myricetin a significant increase in molecular ellipticity compared to the control sample (Rawel *et al.*, 2002a). An increase in molecular ellipticity was also obtained by the derivatization of β -lactoglobulin with quercetin and rutin (Rawel *et al.*, 2003). These changes in spectra highlight some conformational changes induced by the derivatization, affecting the tertiary structure and perturbing the initial conformation of the protein. Chemical modification (acylation and sulfamidation) also produces a similar destructuring of proteins (Gerbanowski *et al.*, 1999).

An intrinsic fluorescence study was also performed e.g. to evaluate changes in tertiary structure caused by reaction of BSA with chlorogenic acid (Rawel *et al.*, 2002b). The maximum emission decreased in intensity as a function of the level of BSA modification by the covalent attachment of the chlorogenic acid and a red shift of the maximum emission was observed (Rawel *et al.*, 2002b). The interaction of the aromatic ring of caffeic acid with aromatic residues such as tyrosine and tryptophan of the protein may be responsible for quenching of fluorescence intensity based on tryptophan fluorescence experiments (Suryaprakash *et al.*, 2000). Such interactions can be confirmed as shown by the progressive decrease in the amount of free tryptophan (Suryaprakash *et al.*, 2000; Rawel *et al.*, 2002b). From the progressive quenching and the red shift observed in the maximum fluorescence emission of derivatized BSA, it can be deduced that conformational changes induced by the modification lead to a further exposure of tryptophan residues to the polar solvent (Jackman & Yada, 1989; Gerbanowski *et al.*, 1999). Similar observations were also recorded for the reaction of β -lactoglobulin with quercetin and rutin (Rawel *et al.*, 2003).

Differential scanning calorimetry was applied to assess the response of BSA-chlorogenic acid derivatives from a thermal stability perspective (Rawel *et al.*, 2002b). The thermal stability of BSA, as reflected by denaturation temperature (T_d), decreased significantly as a function of the level of derivatization (Rawel *et al.*, 2002b). Generally the enthalpy of denaturation (ΔH) of derivatized BSA was also lowered (Rawel *et al.*, 2002b). This decrease was not continuous making conformational changes apparent. The end-product of such a denaturation is generally a partly random coiled protein; this result is caused primarily by the disruption of non-covalent interactions (Ismond *et al.*, 1985). The increase in the random coiled or unordered structure has been documented by CD analysis as described above (Rawel *et al.*, 2002b).

The influence on surface hydrophobicity of BSA molecules by covalent linkage of chlorogenic acid was documented by binding of hydrophobic fluorescence probe, 1-anilino-8-naphthalensulfonate (ANS) (Rawel *et al.*, 2002b). ANS is widely used to monitor conformational changes in proteins and to characterize surface exposure of hydrophobic sites (Hayakawa & Nakai, 1985; Ismond *et al.*, 1985; Ismond *et al.*, 1988; Gerbanowski *et al.*, 1999). The surface hydrophobicity decreased with increasing degree of derivatization of BSA as shown by the low initial

slopes (S_0 -hydrophobicity index), calculated from the corresponding fluorescence intensity vs. protein concentration plots (Rawel *et al.*, 2002b). This means that the surface of BSA becomes hydrophilic. This also explains the better solubility of the BSA-chlorogenic acid derivatives as compared to e.g. myoglobin-phenol derivatives, which exhibited increased hydrophobicity (Kroll *et al.*, 2000; Kroll & Rawel, 2001). According to Hayakawa and Nakai (1985), the decrease in solubility correlates generally well with an increase in surface hydrophobicity as measured by the hydrophobic fluorescence probe ANS. The surface hydrophobicity was also decreased when β -lactoglobulin was derivatized by quercetin and rutin respectively (Rawel *et al.*, 2003). The observed decrease in the surface hydrophobicity can be explained as combined effect of a conformational change of the protein that increased the exposure of some additional hydrophilic regions which were previously buried, the covalent blocking of exposed hydrophobic residues like tryptophan and that the phenolic compounds bound, introduced new hydrophilic components (hydroxyl- and carboxyl groups).

In summary, CD studies and estimation of changes in secondary structures clearly demonstrates that the reaction of proteins with phenolic compounds leads to changes in α -helix. The intrinsic fluorescence, DSC measurements and ANS binding experiments further indicate significant changes in the conformation of protein derivatized with the phenolic compounds.

4. Effect of protein derivatization on the proteolytic *in vitro* digestion

The enzymatic hydrolysis of derivatized food proteins as tested by trypsin, chymotrypsin, pepsin and pancreatin were differently influenced. Exemplary, an increase in concentration of phenolic substance present during derivatization caused the proteolytic digestion to become negatively influenced as reported for myoglobin derivatives (Kroll *et al.*, 2000). Similarly, an alkaline pH lead to a higher derivatization and the proteolytic digestion (e.g. peptic hydrolysis of lysozyme derivatives with chlorogenic acid) was correspondingly poor (Rawel *et al.*, 2000a). Quinic acid (a non aromatic substance) reacts only to small extent with protein, where as starting from resorcinol, ferulic acid, chlorogenic-, caffeic acid, pyrocatechol, hydroquinone, *p*-quinone and gallic acid a corresponding stronger reactivity and a corresponding stronger negative effect on proteolytic digestion was observed. The effect of derivatization of myoglobin and whey proteins with some of these phenolic compounds on all of the four proteolytic digestions tested, were negatively effected (Kroll *et al.*, 2000; Kroll & Rawel, 2001; Rawel *et al.*, 2001a). The results also showed that lysozyme derivatization with these plant phenols leads generally to a better digestion by trypsin, chymotrypsin and pancreatin, where as the peptic digestion is not favoured (Rawel *et al.*, 2001b). With regard to the derivatization of soy glycinin with selected flavonoids, general observations were that the tryptic, chymotryptic and pancreatic hydrolysis were promoted (except that flavone had little or no effect), whereas the peptic digestion remained more or less unaffected (Rawel *et al.*, 2002a). A similar observation was also made, when digesting myoglobin derivatized with the these flavonoids, except here even the peptic digestion was also promoted. Exemplary, the chymotryptic digestion is presented in Fig. 15.

In conclusion, the proteolytic degradation of food proteins

derivatized with phenolic compounds was effected by the pH milieu and the concentration of the phenolic substance present during the derivatization as well as from the phenol and protein configuration.

5. Reactions of phenolics and related substances with enzymes

The plant phenolic substances react with enzymes influencing their physicochemical and structural properties as described above and consequently their *in vitro* enzymatic activity was also adversely affected (Rawel *et al.*, 2000b; Rawel *et al.*, 2001b; Rohn *et al.*, 2001; Rohn *et al.*, 2002a,b). The proteolytic digestion of food proteins with α -chymotrypsin derivatized by phenolic compounds was adversely affected as reported in Rohn *et al.* (2001). This decrease depended on the reactivity of the phenolic and related substances tested as well as on the kind of substrate applied (Rohn *et al.*, 2001). Reactions of phenolic compounds with other enzymes (trypsin, lysozyme, α -amylase) showed similar results with regard to changes in physicochemical properties and the effect on their activities (Rohn *et al.*, 2002b). The evaluation of the lytic activity against gram-positive bacteria (*Micrococcus lysodeikticus*) showed that the lytic degradation of their cell walls by lysozyme derivatives of different phenolic compounds was significantly reduced due to the covalent attachment of the phenolic substances (Rawel *et al.*, 2001b; Rohn *et al.*, 2002b). The soy trypsin inhibitor activity against tryptic digestion declined as tested for the two different substrates (N_α -benzoyl-arginoanilide-BAPA and casein) (Rawel *et al.*, 2002a). Soy trypsin inhibitors cause growth inhibition, pancreatic hypertrophy, and hyperplasia in experimental animals (Sessa *et al.*, 1988). Such biological properties may also be effected by phenolic compounds, but remain to be researched. Figure 16 summarises the detrimental effects on biological activity of the tested enzymes and inhibitor. The standard substrates were applied: For α -amy-

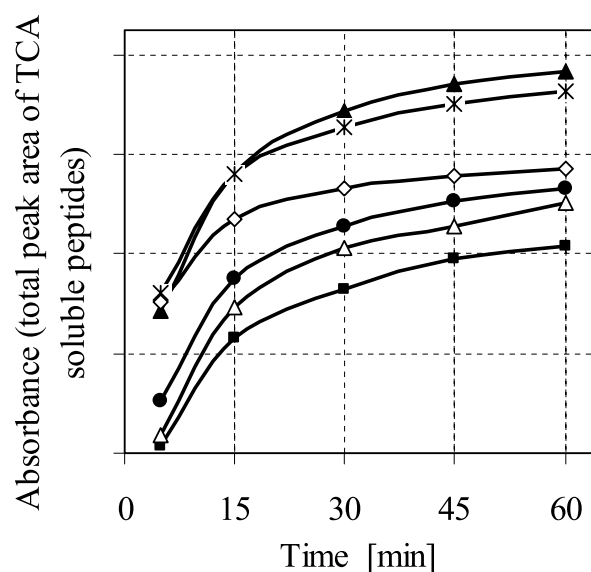


Fig. 15. Chymotryptic digestion of the myoglobin-flavonoid derivatives (1.07 mmol phenol/g protein; Enzyme:Substrate ratio-E : S=1 : 60). Code: 1=■ unmodified (control) myoglobin; 2=△ flavone derivative; 3=● apigenin derivative; 4=▲ kaempferol derivative; 5=* quercetin derivative; 6=◇ myricetin derivative.

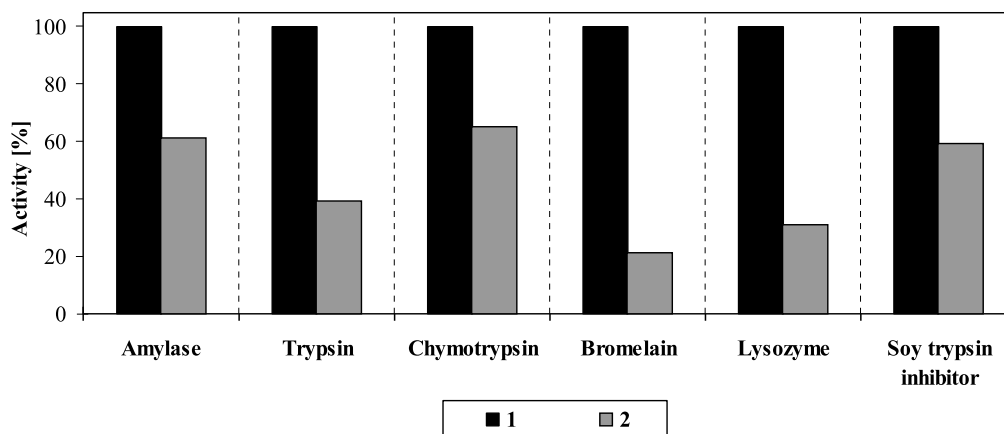


Fig. 16. Effect of derivatization with chlorogenic acid on the biological activity of proteins. Code: 1=control enzyme; 2=enzyme derivatized in presence of 0.28 mmol chlorogenic acid/g protein, except for lysozyme and soy trypsin inhibitor, where 0.14 mmol and 0.3 mmol chlorogenic acid respectively were applied.

lase-potato starch; trypsin-BAEE (N_α -benzoyl-arg-ethyl ester); α -chymotrypsin-N-suc-ala-ala-pro-phe-*p*-nitroanilide; bromelain-N-CBZ-LNPE (N_α -carbobenzoxy-lys-*p*-nitrophenyl ester); lysozyme-cells from *Micrococcus lysodeikticus*; soy trypsin inhibitor- N_α -benzoyl-arg-nitroanilide. Detailed experimental conditions are reported elsewhere (Rawel *et al.*, 2001b; Rohn *et al.*, 2001; Rohn *et al.*, 2002b; Rawel *et al.*, 2002a).

The kinetics of the enzymatic reactions showed that the hydrolysis of selected food proteins becomes slower and the affinity of the enzyme to these substrates declined as measured by Michaelis-Menten constant and maximum velocity of the enzymatic reaction (Rohn *et al.*, 2003). This enzyme inhibition depended on the reactivity of the phenolic and related substances tested as well as on the degree of the derivatization. Further, influence of the enzyme-substrate ratio was also demonstrated. The effects of the derivatization are more pronounced with increasing concentration of the substrates.

6. Conclusions: Nutritional-physiological consequences of such reactions

These interactions of flavonoids with myoglobin confirm our former results with other proteins/enzymes, showing that the prerequisite condition for a phenol to form covalent bonds with proteins lies in its capability to form corresponding quinones or semi-quinone radicals. The later being in position to undergo nucleophilic addition at the amino side chains of the proteins. The reaction proceeds in a further stage, where polymerization may take place.

The nutritional value of food proteins is governed by amino acid composition, ratios of essential amino acids, susceptibility to hydrolysis during digestion, source, and the effects of processing. To optimize the biological utilization of proteins, especially of low-quality proteins in underdeveloped countries, a better understanding is needed of the various interrelated parameters that influence their nutritional value. Enzyme-catalyzed browning reactions of amino acid and proteins with oxidized plant phenols may cause deterioration of food during storage and processing leading to a loss in nutritional quality, which is especially serious in underprivileged countries. Some complex polyphenols

such as tannins are present in many cereal foods, e.g., grain sorghum, the latter being widely used in many African countries. A similar problem also encountered in the utilization of proteins from sunflower, safflower, cotton seed and alfalfa leaf is that they contain high levels of phenolic compounds (Hurrell *et al.*, 1982; Hurrell & Finot, 1984; Matheis & Whitaker, 1984a). Further, alkaline processing of proteins may also be applied in technological treatments of foods and feeds by solubilization and purification, to destroy toxic contaminants, to obtain functional properties, including the formation of textured vegetable protein fibers (Hurrell *et al.*, 1982). These conditions would promote reactions of proteins with secondary plant metabolites as shown above inducing changes in their physico-chemical properties. The enzymatic digestion of the food protein derivatives by trypsin, chymotrypsin, pepsin and pancreatin is adversely effected. And, finally the biological function e.g. enzyme activity could also be negatively influenced.

Further, after intake of food materials containing phenolic compounds and after their corresponding absorption, these free ligands may interact with many transport proteins and enzymes as shown by the inhibition of human spleen tyrosine kinases (Lazaro *et al.*, 1995). Further, phenols have also been attributed with positive properties like having antimutagenic, antimicrobial and anticarcinogenic effects as well as being antioxidants—a possible mechanism by which dietary components protect the body from free radicals and reactive oxygen species (Robards *et al.*, 1999). This is biologically relevant especially with respect to possible toxicological risks of these supposed beneficial antioxidants, apparently able to be converted to electrophilic pro-oxidants (Awad *et al.*, 2000; Boersma *et al.*, 2000; Awad *et al.*, 2001; Awad *et al.*, 2002a,b). The *p*-quinone methide of quercetin may prove to be an important toxic metabolite involved in the cytotoxicity and pro-oxidant effects of quercetin (Boersma *et al.*, 2000). Further, this is of utmost importance, in connection with the recent trends of introducing more and more supplementary foods on the commercial level with the intention of marketing the positive (antioxidative!) effects of flavonoids. These supplementary foods contain high concentrations of flavonoids, which are physiologically disquieting, especially since depending on

the concentration and hydroxyl substituent pattern, they can also act as pro-oxidants (Awad *et al.*, 2000; Boersma *et al.*, 2000; Awad *et al.*, 2001; Awad *et al.*, 2002a,b). Recently, the quercetin adduct formation with glutathione was indicated to be reversible, leading to possibilities for release and further electrophilic reactions of the quercetin quinone methide at cellular sites different from those of its generation (Boersma *et al.*, 2000).

The different aspects of consumption of protein-phenol derivatives and their consequent proteolytic digestion have been adequately commented by Friedman (Friedman, 1997) as follows: "the nutritional and toxicological significance of consuming such protein derivatives is largely unknown."

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